Fiscal Year:	FY 2023	**Task Last Updated:**	FY 04/29/2023
PI Name:	Shelhamer, Mark Sc.D.		
Project Title:	Investigation of Partial-g Effects on Ocular Alignment		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	410-614-5898
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PI Web Page:	https://		
City:	Baltimore	**State:**	MD
Zip Code:	21205-1832	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2019 HERO 80JSC019N0001-FLAGSHIP & OMNIBUS: Human Research Program Crew Health. Appendix A&B
Start Date:	07/28/2020	**End Date:**	07/27/2024
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 07/27/2024 per NSSC information (Ed., 3/20/23)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Schubert, Michael Ph.D. (Johns Hopkins University)		
Grant/Contract No.:	80NSSC20K1498		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This project will provide information on binocular alignment as a measure of otolith asymmetry – more specifically as a measure of the neural compensation for asymmetry, which changes as a function of g level. This low-level function is easily and rapidly measured, and has been validated in vestibular patients and parabolic flight. The project draws on related Human Research Program (HRP) initiatives: Sensorimotor Assessment and Rehabilitation Apparatus (NNX10AO19G, 2010-2014) and Assessment of Otolith Function and Asymmetry as a Corollary to Critical Sensorimotor Performance in Missions of Various Durations (80NSSC19K0487, 2019-2027). Based on our previous studies, we anticipate a threshold of about 0.3 g, where there is a transition from ocular alignment that prevails in 1 g to that which is normal in 0 g (Karmali et al., J Vestibular Res 16:117-125, 2006). A subsequent model suggests a slightly higher (but not abrupt) transition at about 0.6 g (Beaton et al., Frontiers Syst Neurosci 9, 2015); thus, we predict a switching threshold in the range of 0.3 to 0.6 g.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The ability to evaluate vestibular (otolith) function, via simple and rapid testing with minimal apparatus, is useful in a variety of clinical settings. Clinical evaluation of our device and procedures is ongoing, for application to vestibular patients and military personnel with blast injuries.
Task Progress:	With the project now in the implementation phase, preparations are underway for parabolic flights in June 2023. All NASA Institutional Review Board (IRB) approvals have been obtained. The protocol had been on hold with the NASA IRB per guidance from NASA Human Research Program (HRP), until the NASA protocol template and consent form were finalized. This approval was granted in July 2022. A Reliance Acknowledgement is in place, whereby the Johns Hopkins University (JHU) IRB cedes authority to the NASA IRB. Logistics, planning, and documentation have been carried out for Novespace. Test subjects, provided by Novespace, will be shared with Gary Strangman's project. These subjects have parabolic-flight experience. This will partly mimic the spaceflight situation that is relevant to this project: astronauts on their way to the moon (Mars), who have adapted to 0g during transit, and who are then subjected to altered g during and after lunar (Martian) landing. Most notably, our alignment testing has been implemented on a Meta Quest 2 virtual-reality headset. An Xbox hand controller is used by the subject to interact with the headset; modifications have been made to the controllers to remove unnecessary (and distracting) controls in order to simplify operation in flight. Extensive validation, verification, and familiarization are now proceeding. A single parabolic flight with Zero-G Corporation is planned for May, in order to verify the operation of the headset in altered gravity (planning for this flight has been beset by scheduling and maintenance issues). (A significant concern is the stability of the visual images on the VR display under altered gravity levels. This might be adversely impacted if the system uses inertial sensors (accelerometers) to maintain stability of the rendered visual scene as the head moves. Efforts to resolve this have been aided by discussions with Laurence Harris and Pascal Serrarens, who have experience with the Quest platform in parabolic flight.)
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Articles in Peer-reviewed Journals	Macaulay TR, Wood SJ, Bollinger A, Schubert MC, Shelhamer M, Bishop MO, Reschke MF, Clément G. "Comparison of asymmetry between perceptual, ocular, and postural vestibular screening tests." Brain Sci. 2023 Jan 23;13(2):189. https://doi.org/10.3390/brainsci13020189 ; PubMed PMID: 36831732; PubMed Central PMCID: PMC9954277 , Jan-2023